Laboratory Work 4 Activity | Forecasting Time-Series Data in Power BI

Patrick Jay Cabusas
Joseph Vistal
Instructor

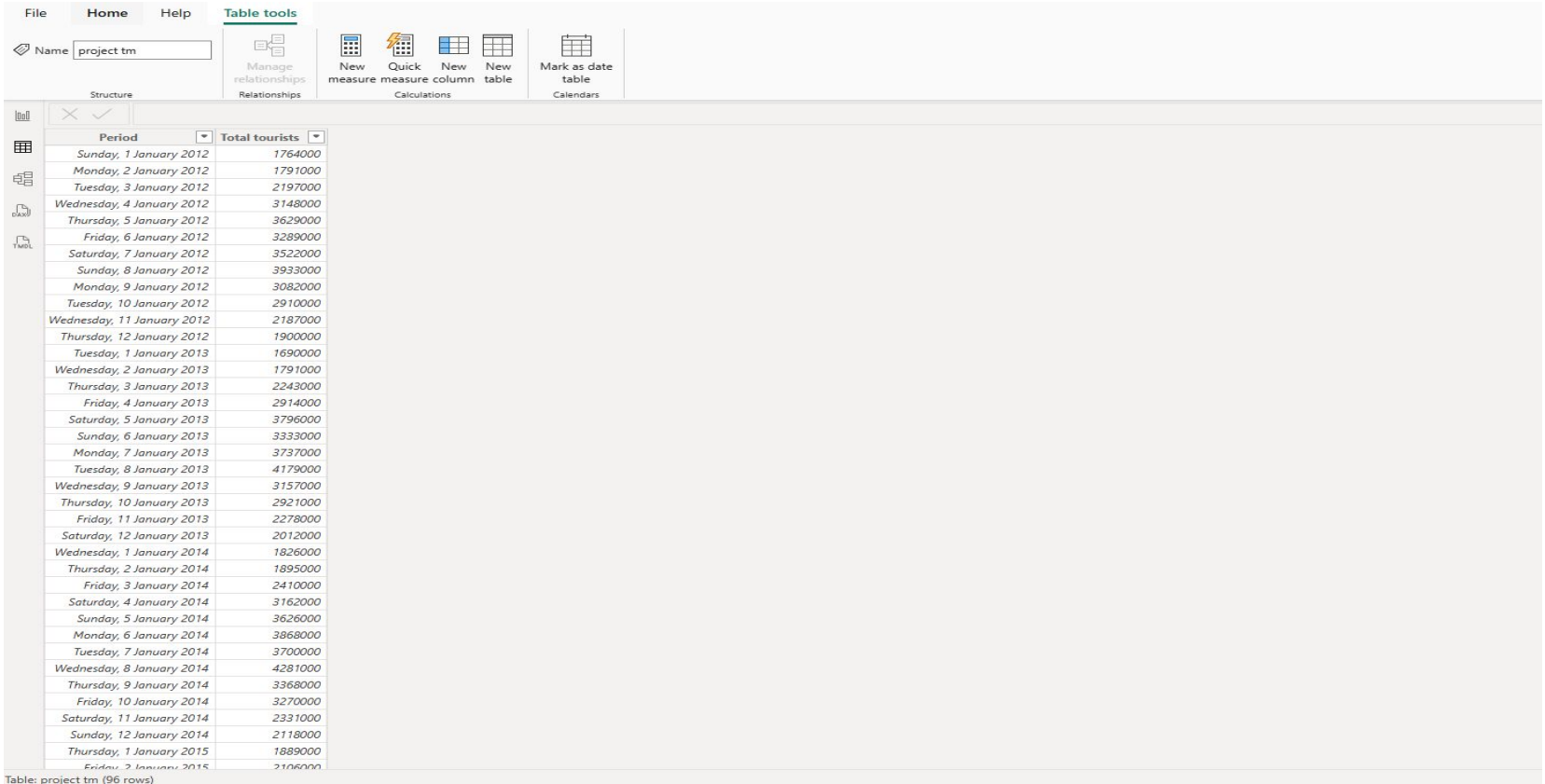
Step 1: Download the Sample Dataset

project.tm.txt

File Edit View

```
Period,Total tourists
1/1/2012,1764000
2/1/2012,1791000
3/1/2012,2197000
4/1/2012,3148000
5/1/2012,3629000
6/1/2012,3289000
7/1/2012,3522000
8/1/2012,3933000
9/1/2012,3082000
10/1/2012,2910000
11/1/2012,2187000
12/1/2012,1900000
1/1/2013,1690000
2/1/2013,1791000
3/1/2013,2243000
4/1/2013,2914000
5/1/2013,3796000
6/1/2013,3333000
7/1/2013,3737000
8/1/2013,4179000
9/1/2013,3157000
10/1/2013,2921000
11/1/2013,2278000
12/1/2013,2012000
1/1/2014,1826000
2/1/2014,1895000
3/1/2014,2410000
4/1/2014,3162000
5/1/2014,3626000
6/1/2014,3868000
7/1/2014,3700000
8/1/2014,4281000
9/1/2014,3368000
10/1/2014,3270000
11/1/2014,2331000
12/1/2014,2118000
1/1/2015,1889000
2/1/2015,2106000
3/1/2015,2373000
4/1/2015,3272000
5/1/2015,4036000
6/1/2015,3617000
7/1/2015,4079000
8/1/2015,4449000
9/1/2015,3316000
10/1/2015,3401000
11/1/2015,2522000
12/1/2015,2258000
1/1/2016,2048000
2/1/2016,2324000
```

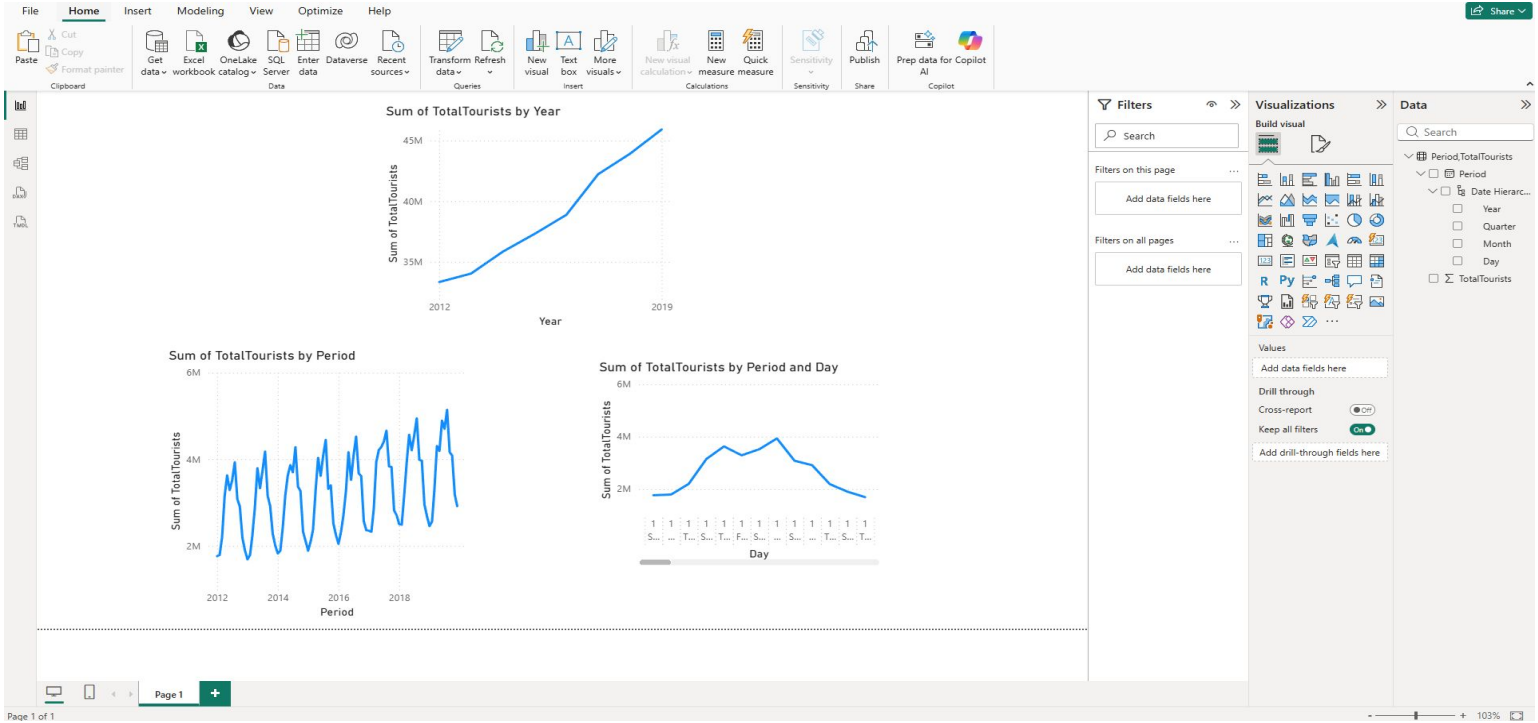
Step 2: Import Data into Power BI



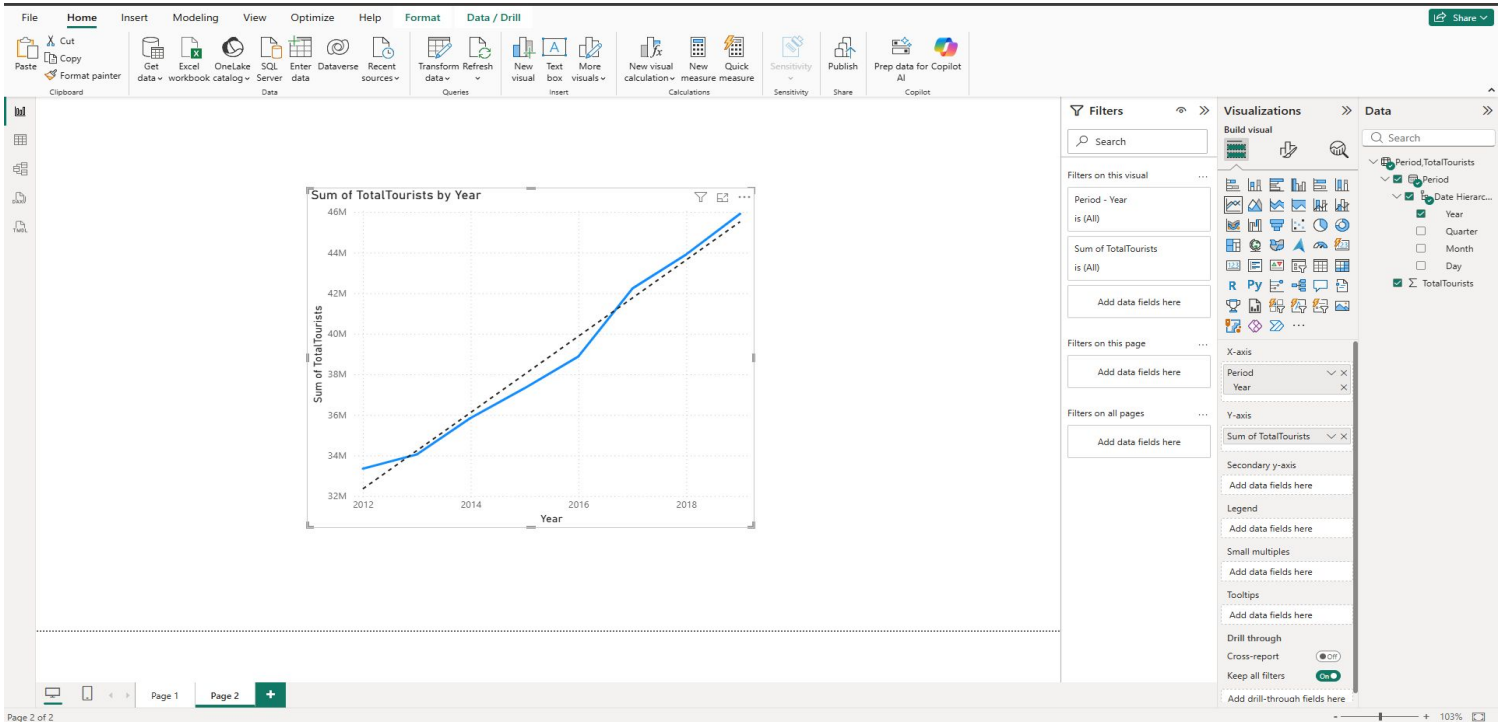
Period	Total tourists
Sunday, 1 January 2012	1764000
Monday, 2 January 2012	1791000
Tuesday, 3 January 2012	2197000
Wednesday, 4 January 2012	3148000
Thursday, 5 January 2012	3629000
Friday, 6 January 2012	3289000
Saturday, 7 January 2012	3522000
Sunday, 8 January 2012	3933000
Monday, 9 January 2012	3082000
Tuesday, 10 January 2012	2910000
Wednesday, 11 January 2012	2187000
Thursday, 12 January 2012	1900000
Tuesday, 1 January 2013	1690000
Wednesday, 2 January 2013	1791000
Thursday, 3 January 2013	2243000
Friday, 4 January 2013	2914000
Saturday, 5 January 2013	3796000
Sunday, 6 January 2013	3333000
Monday, 7 January 2013	3737000
Tuesday, 8 January 2013	4179000
Wednesday, 9 January 2013	3157000
Thursday, 10 January 2013	2921000
Friday, 11 January 2013	2278000
Saturday, 12 January 2013	2012000
Wednesday, 1 January 2014	1826000
Thursday, 2 January 2014	1895000
Friday, 3 January 2014	2410000
Saturday, 4 January 2014	3162000
Sunday, 5 January 2014	3626000
Monday, 6 January 2014	3868000
Tuesday, 7 January 2014	3700000
Wednesday, 8 January 2014	4281000
Thursday, 9 January 2014	3368000
Friday, 10 January 2014	3270000
Saturday, 11 January 2014	2331000
Sunday, 12 January 2014	2118000
Thursday, 1 January 2015	1889000
Friday, 2 January 2015	2106000

Table: project tm (96 rows)

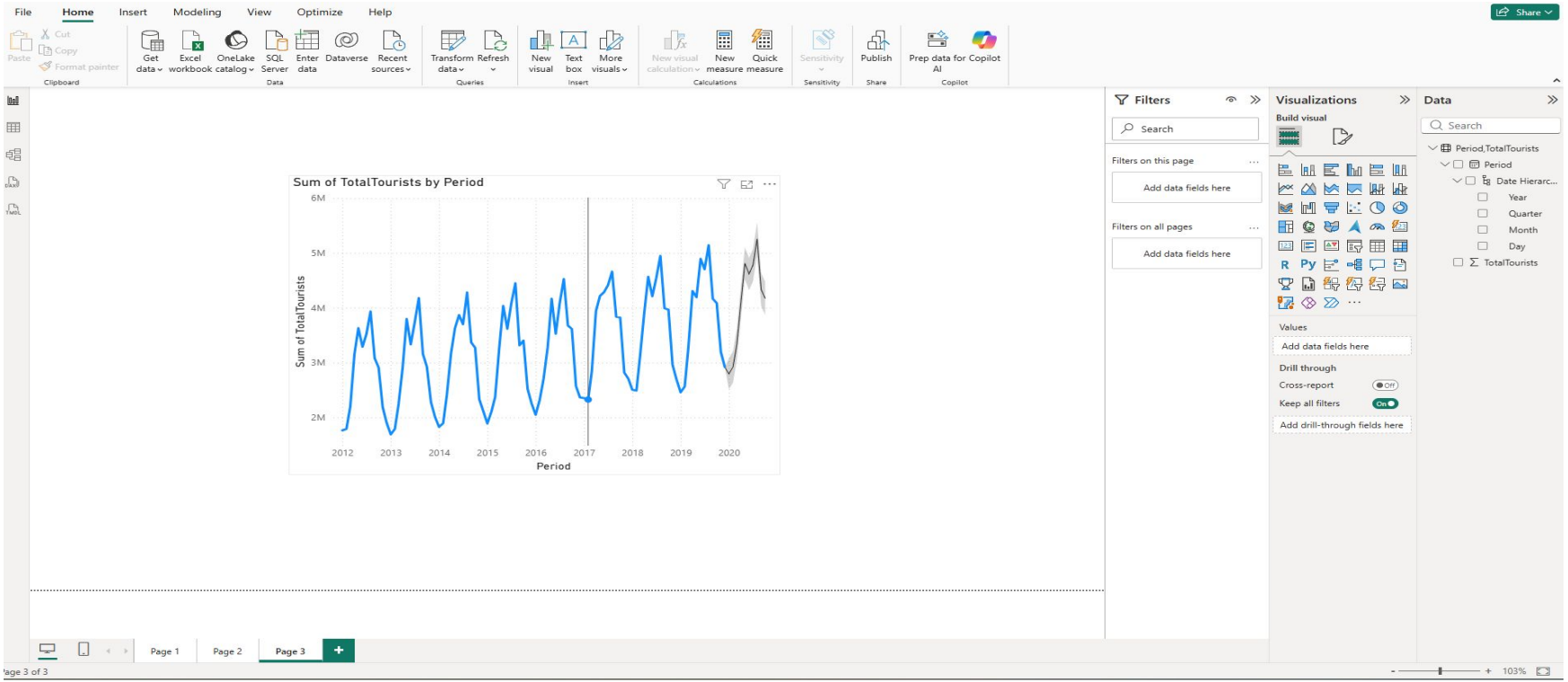
Step 3: Create a Line Chart for Yearly Data



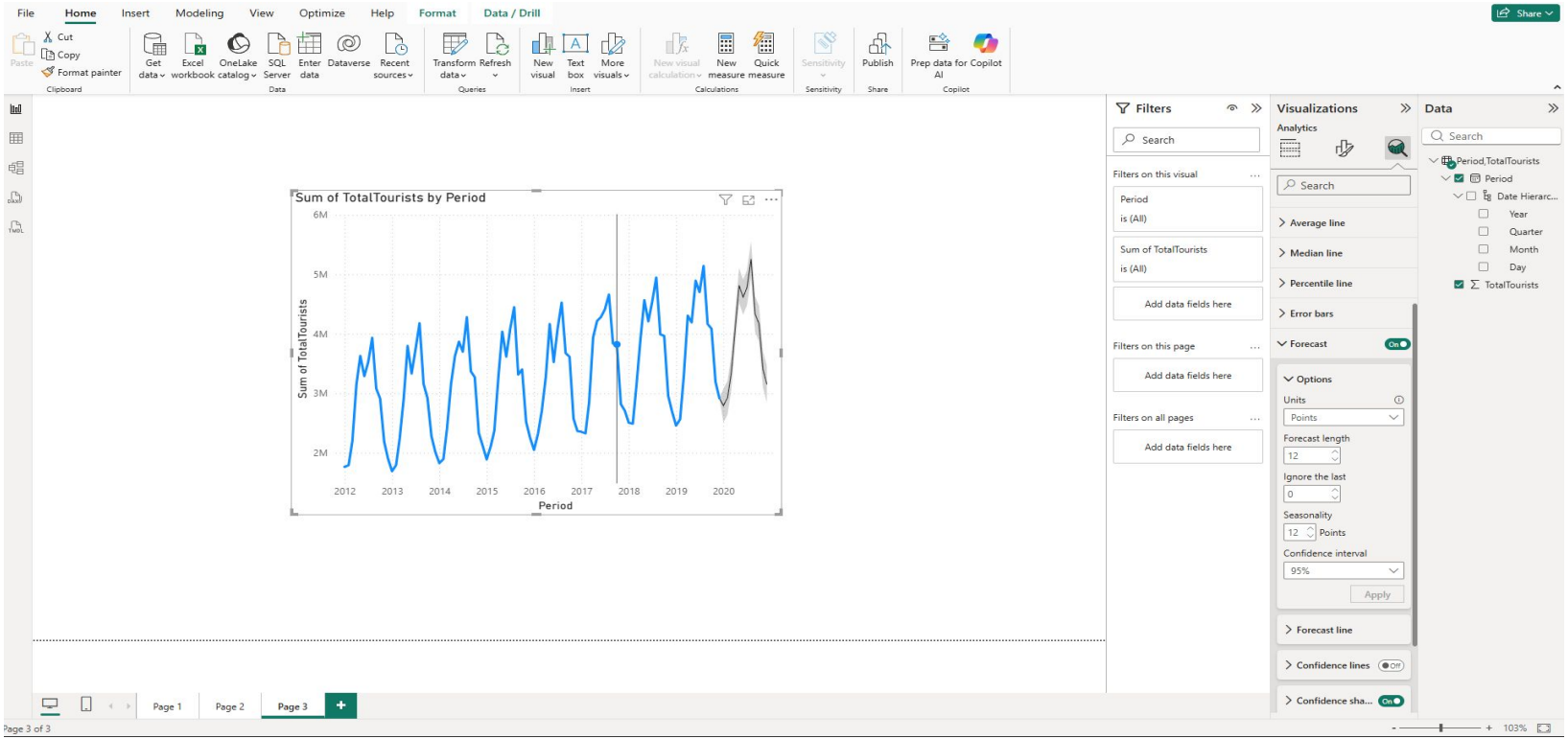
Step 4: Add a Trend Line (Optional but Recommended)



Step 5: Add Forecasting



Step 6: Configure the Forecast



7: Analyze the Forecast

1. Observe the generated forecast for the year 2020.

- The forecast for 2020 The gray shaded area at the far right is the forecast. It predicts tourism would continue its seasonal pattern and potentially reach around 5M–6M tourists, following the upward trend from previous years.

2. Note the confidence interval (the shaded gray area). What does its narrowness

suggest about the forecast's accuracy for this period?

- The shaded gray band is relatively narrow, which means the model is highly confident in its prediction. This makes sense because the data up to 2019 showed a very consistent, regular seasonal pattern — the model could clearly see the repeating cycle and projected it forward with high certainty.

3. Compare the forecast to the actual data for 2019.

- Compare forecast vs actual 2019 data Looking at 2019, the actual data closely follows the same seasonal wave pattern that the model learned from. The peak months (around mid-year) hit ~5M, and the model correctly captured this rhythm — which is why it confidently projects the same shape into 2020.

Part 2: Enhancement - Forecasting with Imperfect Data
Step 8: Prepare a New Dataset (or use a modified version)

Period	TotalTourists
Wednesday, April 1, 2020	50000
Friday, May 1, 2020	45000
Monday, June 1, 2020	120000
Wednesday, July 1, 2020	300000
Saturday, August 1, 2020	450000
Tuesday, September 1, 2020	380000
Thursday, October 1, 2020	420000
Sunday, November 1, 2020	390000
Tuesday, December 1, 2020	410000
Friday, January 1, 2021	380000
Monday, February 1, 2021	400000
Monday, March 1, 2021	650000
Thursday, April 1, 2021	900000
Saturday, May 1, 2021	1100000
Tuesday, June 1, 2021	1300000
Thursday, July 1, 2021	1800000
Sunday, August 1, 2021	2100000
Wednesday, September 1, 202	1900000
Friday, October 1, 2021	2000000
Monday, November 1, 2021	1850000
Wednesday, December 1, 202	1950000
Saturday, January 1, 2022	2100000
Tuesday, February 1, 2022	2300000

Period	TotalTourists
Tuesday, March 1, 2022	2800000
Friday, April 1, 2022	3200000
Sunday, May 1, 2022	3500000
Wednesday, June 1, 2022	3800000
Friday, July 1, 2022	4100000
Monday, August 1, 2022	4300000
Thursday, September 1, 2022	3900000
Saturday, October 1, 2022	3700000
Tuesday, November 1, 2022	3200000
Thursday, December 1, 2022	3000000
Sunday, January 1, 2023	3100000
Wednesday, February 1, 2023	3300000
Wednesday, March 1, 2023	3700000
Saturday, April 1, 2023	4000000
Monday, May 1, 2023	4200000
Thursday, June 1, 2023	4500000
Saturday, July 1, 2023	4700000
Tuesday, August 1, 2023	4900000
Friday, September 1, 2023	4400000
Sunday, October 1, 2023	4200000
Wednesday, November 1, 202	3800000
Friday, December 1, 2023	3500000
Monday, January 1, 2024	3600000
Thursday, February 1, 2024	3800000

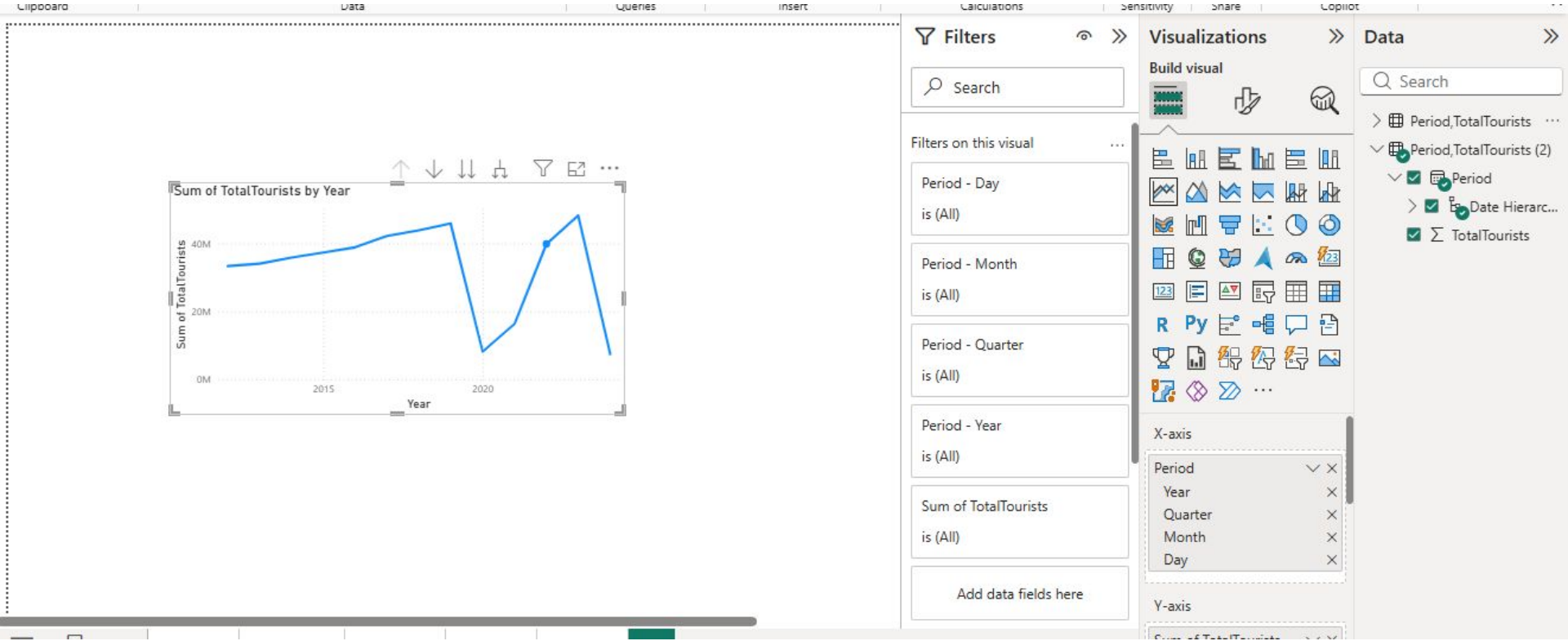
Step 9: Import and Prepare Your New Dataset

	Period	TotalTourists
97	1/1/2020	2654000
98	2/1/2020	2100000
99	3/1/2020	800000
100	4/1/2020	50000
101	5/1/2020	45000
102	6/1/2020	120000
103	7/1/2020	300000
104	8/1/2020	450000
105	9/1/2020	380000
106	10/1/2020	420000
107	11/1/2020	390000
108	12/1/2020	410000
109	1/1/2021	380000
110	2/1/2021	400000
111	3/1/2021	650000
112	4/1/2021	900000
113	5/1/2021	1100000
114	6/1/2021	1300000
115	7/1/2021	1800000
116	8/1/2021	2100000
117	9/1/2021	1900000
118	10/1/2021	2000000
119	11/1/2021	1850000
120	12/1/2021	1950000

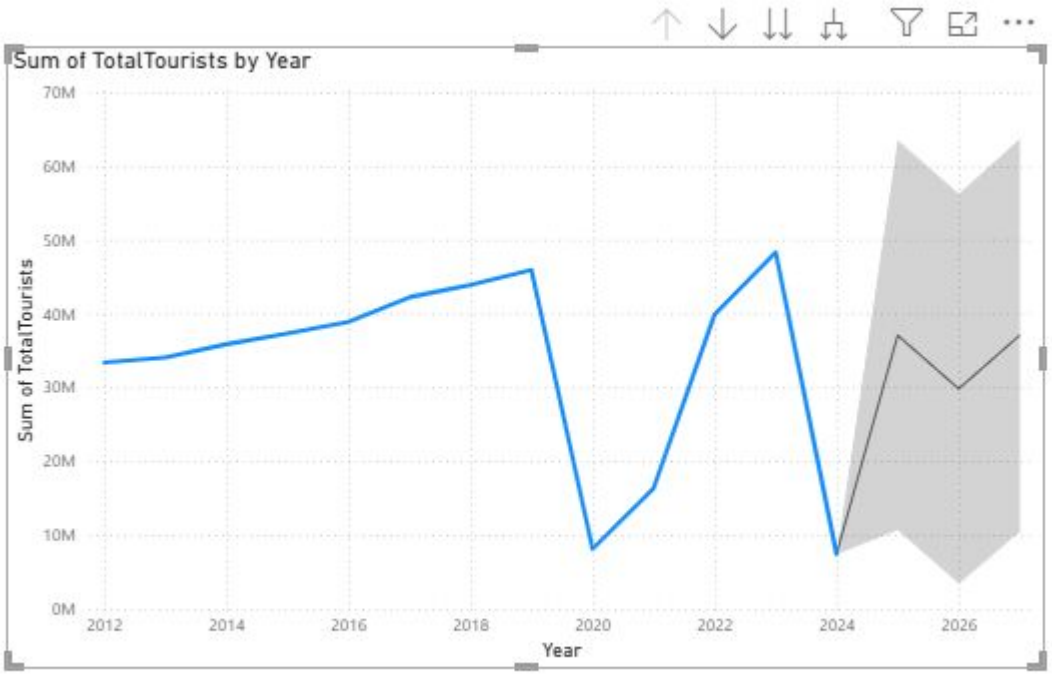
```
= Table.SelectRows(#"Changed Type",
```

	Period	TotalTourists
123	3/1/2022	2800000
124	4/1/2022	3200000
125	5/1/2022	3500000
126	6/1/2022	3800000
127	7/1/2022	4100000
128	8/1/2022	4300000
129	9/1/2022	3900000
130	10/1/2022	3700000
131	11/1/2022	3200000
132	12/1/2022	3000000
133	1/1/2023	3100000
134	2/1/2023	3300000
135	3/1/2023	3700000
136	4/1/2023	4000000
137	5/1/2023	4200000
138	6/1/2023	4500000
139	7/1/2023	4700000
140	8/1/2023	4900000
141	9/1/2023	4400000
142	10/1/2023	4200000
143	11/1/2023	3800000
144	12/1/2023	3500000
145	1/1/2024	3600000
146	2/1/2024	3800000

Step 10: Create a Line Chart for Your Data



Step 11: Add and Configure the Forecast

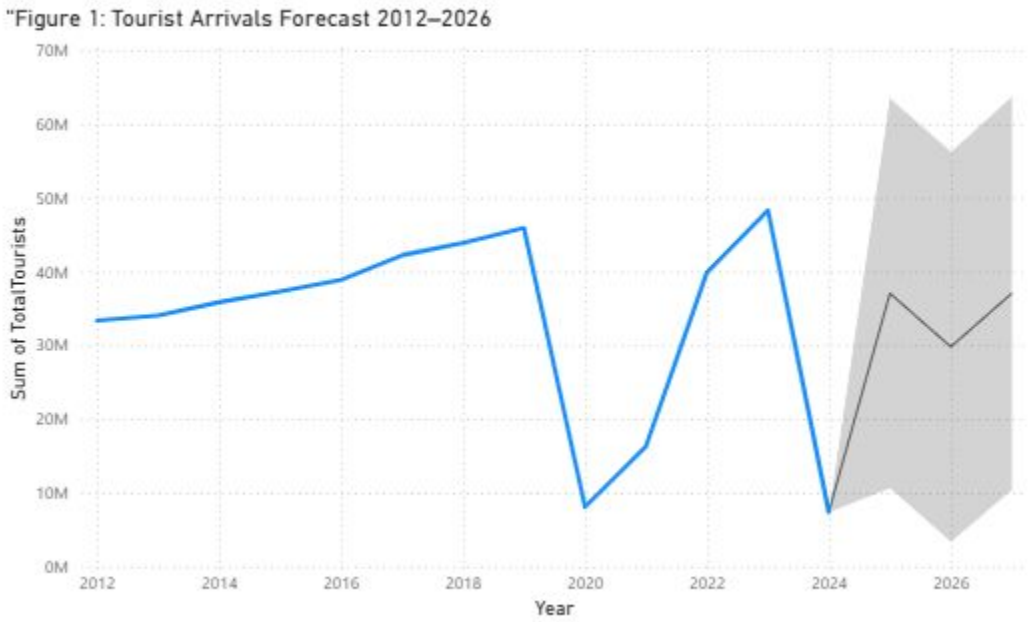


Step 12: Analyze and Report Your Findings

1. Describe your dataset: Briefly explain what data you used, its source, and the time
period covered.

The dataset contains monthly international tourist arrivals spanning over 12 years. It includes the pre-COVID growth period, the dramatic disruption of 2020, and the subsequent recovery from 2021 to 2024. Additional data points were added to simulate real-world disruptions and recovery patterns.

2. Visualize your forecast: Include a screenshot of your Power BI line chart with the forecast applied.



3. Interpret the forecast:

The dataset reveals a clear **upward trend from 2012 to 2019**, with tourist arrivals growing steadily each year, followed by a dramatic collapse in 2020 due to the COVID-19 pandemic, and a strong recovery from 2021 to 2024. The data also shows a consistent **yearly seasonal pattern**, with peak arrivals during summer months and lower numbers during winter. Power BI's forecast predicts **continued growth from 2024 to 2026**, following the recovery momentum of recent years. However, the **confidence interval is noticeably wide**, which directly reflects the uncertainty introduced by the COVID-19 disruption in 2020 — the wider the shaded area, the less certain the model is about future values. When experimenting with the **Ignore Last** setting, ignoring more recent data points produced a flatter and more pessimistic forecast, showing how sensitive the model is to which data is included. Similarly, raising the **confidence interval to 99%** widened the shaded area significantly, while lowering it to **90%** made it narrower but riskier. Overall, while Power BI's forecast follows a reasonable growth trajectory based on past patterns, the wide confidence band suggests that the forecast should be interpreted with caution given the significant disruption in the historical data.

4. Conclusion: Based on your analysis, how useful is Power BI's forecasting feature for your chosen dataset? What are its strengths and limitations in this context?

Power BI's forecasting feature is a **powerful and accessible tool** for analyzing tourism trends. It works exceptionally well for data with consistent patterns and gradual changes. However its reliability is significantly reduced when historical data contains major disruptions like the COVID-19 pandemic. For datasets like this one, Power BI forecasting is best used as a **starting point for analysis** rather than a definitive prediction tool. It should always be combined with **domain knowledge and human judgment** for the most accurate interpretation.